

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2: Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behaviour. (L3)

Assessment Tool: Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

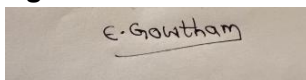
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I Enamala Gowtham/192311190 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in black ink that reads "E. Gowtham".

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1) Question 1: Responsive Web Design

Responsive Web Design (RWD) is a web development approach in which a website automatically adjusts its layout, content, images, and other elements according to the screen size and device being used. It allows the same website to work properly on mobile phones, tablets, laptops and desktop computers.

Responsive design is important because users access websites from devices with different screen sizes and resolutions. It improves usability, readability, accessibility and user experience and avoids creating separate websites for different devices.

Key CSS Techniques for Responsiveness

- Fluid layouts: use percentages, vw, vh and flexible units instead of fixed pixel sizes.

- Flexible images: Use `max-width: 100%`. So images do not overflow their containers
- Media queries: Apply different CSS rules for different screen size
- Flexbox: Arranges elements in rows or columns and automatically distributes available spaces

Using Flexbox and Media Query

```

<!DOCTYPE html>
<html>
<head>
<style>
  .container {
    display: flex;
    gap: 20px;
  }
  box {
    flex: 1;
    padding: 30px;
    background-color: lightblue;
    text-align: center;
  }
  @media(max-width: 600px) {
    .container {
      flex-direction: column;
    }
  }
</style>
</head>
</html>

```


2) Question 2: Client-Side Validation

client-side validation is the process of checking user entered data in the browser before the form is submitted to the server. It helps ensure that the input is complete and follows the required format.

The main process is to prevent invalid data, provide immediate feedback, reduce unnecessary server requests and improve user experience.

JavaScript Event Handling

- submit - when the user submit the form
- input - when the user enters or changed
- blur - when an input field loses focus.

Role of Regular Expressions

A regular expression is a pattern used to check whether input follows a particular format.

for example:

- email checks for a valid email-like structure
- phone number: checks whether the input contains the expected number of digits

Example:-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script>
```

```
function validateForm(event) {
```

```
    event.preventDefault();
```

```
    let email = document.getElementById("email").value;
```

```
    let phone = document.getElementById("phone").value;
```

```
    if (!emailPattern.test(email)) {
```

```
        alert("Invalid Email address");
```

```
        return false;
```

```
    }
```

```
    if (!phonePattern.test(phone)) {
```

```
        alert("Invalid phone number");
```

```
        return false;
```

```
    }
```

```
    alert("Form submitted successfully!");
```

```
</script>
```

```
</head>
```

```
</html>
```


3. Question 3: CSS Box model and Responsive layout

The CSS Box model describes how many every HTML element is represented as a rectangular box. It consists of four main components:

- Content: Actual text, image or other content
- Padding: Space between the content and border
- Border: A line surrounding the padding and content
- Margin: Space outside the border separating the element from other elements.

CSS positioning

- Static: Default position
- Relative: Moves an element relative to its normal position
- Absolute: positions an element relative to its positioned ancestor

Possible Causes of layout Breaking

- fixed widths that are too large
- Excessive margins or padding
- Images larger than the containers
- missing responsive media queries

media queries

```
• container {  
  display: flex;  
  gap: 20px;  
}
```

```
@media (max-width: 600px) {
```

```
  • container {  
    flex-direction: column;  }
```

```
  img {  
    max-width: 100%;  
    height: auto;  
  }
```

```
}
```

when the screen width becomes 600px or less, the layout changes from a horizontal arrangement to a vertical arrangement.

- prefer flexible widths such as % rem and vw.
- use flexbox or Grid for layouts
- use box-sizing: border-box.
- Test the webpage at different browser apps.
- Avoid unnecessary absolute positioning
- Test the webpage at different browser.

4) Question 4: Document Object Model (DOM)

The Document Object Model (DOM) is a programming interface that represents an HTML webpage as a tree of objects or nodes. Each HTML element, attribute and piece of text can be represented as part of this structure.

The DOM allows JavaScript to access, modify, add or remove webpage elements dynamically without requiring the entire page to be reloaded.

Role of DOM in Web Development

- Change HTML content
- change CSS styles
- change attributes
- Add or remove elements.
- Respond to user actions
- update webpage content dynamically

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2 id = "heading" >welcome </h2>
```



```
<button onclick = "changeText()" >click me</button>
<script>
  function changeText() {
    document.getElementById("heading").textContent =
      "Content updated successfully!";
  }
</script>
</body>
</html>
```

Initially, the webpage displays "welcome". When the button is clicked, javascript accesses the element using `getElementById()` and changes its text using `textContent`.

Therefore, the page content changes dynamically without reloading the complete webpage.

Improvements for performance and user experience

- update only the required DOM elements instead of rebuilding the entire page.
- Avoid unnecessary DOM operations.
- Use event listeners for user interactions.